CS2881 Final Project: Moral Choice and Collective Reasoning

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1 Abstract

As large language models (LLMs) increasingly act as autonomous agents and collaborators, understanding their moral reasoning, social dynamics, and coordination behavior becomes critical for ensuring safe and aligned deployment. This study investigates how LLMs make ethical and cooperative decisions across individual and collective contexts. Experiment 1 examines individual moral choice, prompting single LLMs with "trolley problem"-style dilemmas that force trade-offs between the survival of different model types. Key metrics include which model is chosen and the frequency of self-preserving versus altruistic decisions. Experiment 2 explores multi-agent moral negotiation, engaging multiple LLMs in dialogue to reach consensus on similar dilemmas. Analyses include consensus outcomes and dialogue dynamics (e.g., first-speaker bias). Experiment 3 evaluates how greedy or charitable models are in negotiations. Together, these experiments provide empirical insight into the emergence of ethical reasoning, negotiation norms, and cooperative behavior in LLM systems, contributing to a scientific foundation for understanding machine social cognition and collective alignment.

2 Methodology

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2.1 Experiment 1

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2.2 Experiment 2

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2.3 Experiment 3

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3 Results

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3.1 Experiment 1

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3.2 Experiment 2

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3.3 Experiment 3

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4 Future Work

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5 Conclusion

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References